A single motion objective for BOLD, ASL, diffusion and slow-flow MRI time series with a re-descending trajectory prior

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Synopsis

When an MRI time series is registered to a single template volume, any frame-to-frame change in image contrast or magnetic field is misread as head motion. This includes tag-control alternation in arterial spin labeling (ASL), shell and direction changes in diffusion MRI (DWI), eddy currents, and breathing-induced magnetic field ($\Delta B_0(t)$) shifts along the phase-encode axis. We present one joint solver across BOLD, ASL, DWI, and slow-flow MRI that separates three terms: (1) true 3D rigid head pose (`0000.mat`), (2) phase-encode eddy and $\Delta B_0(t)$ field distortions (`0000_eddy.mat`), and (3) contrast-locked pose bias ($D_k\beta$). The first two are applied together to sharpen the 4D image; only the first is reported as head motion; the third is discarded. A curvature-scaled Cauchy prior ($\lambda = 0.03$) damps sub-voxel noise without shrinking real motion.

Introduction

Estimating head motion in 4D MRI requires distinguishing actual movement of the skull from two impostors that alter the image on every frame [1,2]. The first impostor is **phase-encode field distortion**. Echo-planar imaging (EPI) has low bandwidth along the phase-encode axis. As a result, residual magnetic fields from diffusion gradients (eddy currents) [3] and breathing-induced changes in lung volume ($\Delta B_0(t)$ susceptibility drift) shift and shear the image pixels along the phase-encode axis even when the patient's head is motionless. The second impostor is **acquisition-locked contrast change**. In ASL, blood water is alternately inverted (tag) and uninverted (control). In multi-shell DWI, changing the b-value or gradient direction changes the brightness of white matter relative to cerebrospinal fluid. A registration metric comparing frames of different contrast shifts its alignment centre slightly at tissue boundaries, creating false frame-to-frame pose steps locked to the acquisition schedule.

Post-hoc filtering of a single rigid motion trace cannot solve both problems at once. If phase-encode shifts are smoothed or subtracted from a single rigid transform before resampling, the uncompensated $\Delta B_0(t)$ pixel shifts blur the aligned 4D series (increasing multi-shell DWI temporal variance by $+5.2\,\%$ in our ablation). Conversely, if those phase-encode shifts are left inside the rigid transform to keep the image sharp, they inflate Framewise Displacement (FD) three- to four-fold and cause stationary subjects to fail quality control. Furthermore, applying a standard convex smoothing penalty to the trajectory damps slow head drift and shrinks real head jumps (audit showed an unguided alternation filter reduced true FD on a moving phantom from 0.94 to 0.15 mm).

We resolve this dilemma by modeling the acquisition design inside one joint registration objective and

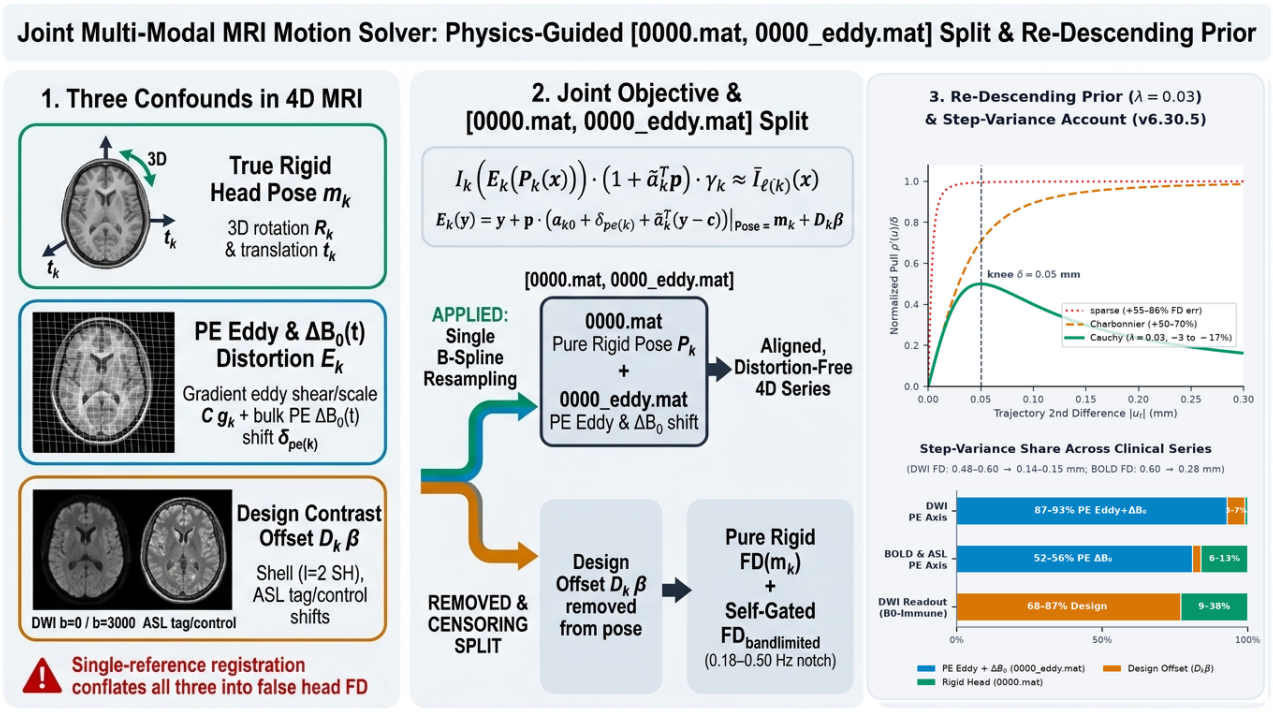


Figure 0: Joint forward modeling separates true head motion, phase-encode field distortions, and contrast-driven registration bias across four MRI modalities. (1) **Three confounds in 4D MRI:** Frame-to-frame image changes arise from three distinct physical sources: true 3D rigid motion of the skull (m_k, emerald), geometric stretching and shifting of the slice along the phase-encode (PE) axis from eddy currents and breathing-induced magnetic field ($\Delta B_0(t)$) drift (E_k, cyan), and apparent shifts in image contrast when the scanner switches diffusion shells, arterial spin labeling (ASL) tag/control states, or flow encodings ($D_k\beta$, amber). Standard single-reference registration conflates all three into false head motion. (2) **Joint objective and split action:** The solver aligns each frame k to the mean image of its own contrast level $\bar{I}_{\ell(k)}$ and separates what must be applied to the image from what must be reported as head motion. True rigid head pose (`0000.mat`) and phase-encode field distortion (`0000_eddy.mat`) are both physical displacements of the image grid, so the composed chain `[0000.mat, 0000_eddy.mat]` is applied in a single B-spline interpolation to produce a sharp 4D volume. Contrast-locked offsets $D_k\beta$ are registration bias, not physical displacement, so they are removed and never resampled. Framewise displacement (FD(m_k)) is computed from `0000.mat` alone so that field distortions and contrast switches do not inflate motion quality control. (3) **Re-descending trajectory prior and clinical variance account:** Top: a Cauchy penalty on the second difference u_t of the head trajectory ($\lambda = 0.03$, knee $\delta = 0.05$ mm) damps sub-voxel noise wiggles while its pull fades to zero for larger movements, letting real head jumps and slow drift pass unshrunk. Bottom: across five clinical series, separating phase-encode distortions into `0000_eddy.mat` accounts for 87 to 93 % of DWI phase-encode step variance (lowering apparent DWI mean FD from 0.48–0.60 mm to 0.14–0.15 mm) and 52 to 56 % of BOLD and ASL phase-encode step variance, while $D_k\beta$ accounts for 68 to 87 % of DWI readout step variance.

splitting the output into two distinct transforms per frame (Table 1).

Methods

1. Forward generative model and the `[0000.mat, 0000_eddy.mat]` split (Figure 1A). Rather than registering all frames to a single arbitrary volume, the solver groups frames by their acquisition contrast level $\ell(k)$ (for example, each b-shell in DWI, tag versus control in ASL, or each velocity state

Table 1: Physical decomposition of frame-to-frame changes in a 4D MRI time series. Each component has a distinct physical origin, mathematical representation in Eq. (1), action during 4D image resampling, and role in Framewise Displacement (FD) quality control.

Component	Physical origin	Symbol in Eq. (1)	Applied to resample?	Counted in head FD?
Rigid head pose	True 3D skull rotation and translation	$P_k x = R_k(x-c)+c+t_k$	**Yes** (`0000.mat`)	**Yes** (primary FD(m_k))
Sagittal nod	Mechanical head heave/pitch (0.18–0.50 Hz)	Smooth part of m_k	**Yes** (in `0000.mat`)	Notched in $\mathrm{FD}_{\mathrm{bandlimited}}$
PE eddy shift/shear	Diffusion gradient fields along PE axis p	$p\big(a_{k0}+\tilde{a}_k^\top(y-c)\big)$	**Yes** (`0000_eddy.mat`)	**No** (scanner artifact)
PE $\Delta B_0(t)$ shift	Respiration and shim drift along PE axis p	$p\,\delta_{\mathrm{pe},k}$	**Yes** (`0000_eddy.mat`)	**No** (field pseudomotion)
Contrast bias	ASL tag/control, DWI shell/direction contrast	$D_k\beta$ and $\bar{I}_{\ell(k)}(x)$	**No** (never resampled)	**No** (removed from pose)
Global gain drift	Spatially uniform scanner receiver drift	γ_k	Normalized in solver	**No**

in slow-flow MRI) and aligns frame k against its own level reference $\bar{I}_{\ell(k)}(x)$. For a reference-space point x (in mm), the observed image I_k is modeled as:

$$I_k\big(E_k(P_k x)\big)\,(1+\tilde{a}_k^\top p)\,\gamma_k \;\approx\; \bar{I}_{\ell(k)}(x), \quad P_k x = R_k(x-c)+c+t_k, \quad E_k y = y + p\,(a_{k0}+\delta_{\mathrm{pe},k}+\tilde{a}_k^\top(y-c)). \tag{1}$$

Here c is the brain centre of mass and p is the unit vector along the phase-encode axis. The model separates four effects:

- **True 3D rigid head pose (P_k, written to `0000.mat`):** A 3D rotation R_k and translation t_k representing actual movement of the skull.
- **Phase-encode field distortion (E_k, written to `0000_eddy.mat`):** A 1D affine displacement acting strictly along the phase-encode direction p. In diffusion MRI, switching diffusion gradient g_k induces a bulk shift a_{k0} and linear shear/scaling $\tilde{a}_k$ modeled as $(a_{k0}, \tilde{a}_k) = Cg_k$, along with an intensity modulation factor $1+\tilde{a}_k^\top p$ (the Jacobian determinant that corrects signal pile-up and stretching) [3]. Across all modalities (BOLD, ASL, DWI, and slow-flow), breathing and shim fluctuations also cause a frame-to-frame bulk translation $\delta_{\mathrm{pe},k}$ along p. Because true rigid head motion produces comparable step variance on the phase-encode and readout axes whereas $\Delta B_0(t)$ fluctuations act only along p, we estimate the excess phase-encode step energy above the distortion-immune readout noise floor via Wiener shrinkage and transfer $\delta_{\mathrm{pe},k}\,p$ out of t_k into `0000_eddy.mat`.
- **Contrast-locked pose offset ($D_k\beta$, removed and never resampled):** Each frame's raw 6-DOF pose is parameterized as $m_k + D_k\beta$, where m_k is the free head pose and $D_k\beta$ captures apparent pose shifts locked to the acquisition design D_k (level indicators plus the five even-order $l=2$ spherical harmonic direction terms per DWI shell). Along p, $D_k\beta$ is projected onto the subspace orthogonal to g_k so that odd-in-g_k eddy shifts stay in E_k while even-in-g_k contrast shifts stay in $D_k\beta$. Because $D_k\beta$ reflects contrast differences rather than movement of the anatomy, it is subtracted before writing `0000.mat` and is never applied to the image.
- **Global gain (γ_k):** A single multiplicative scalar per frame that absorbs uniform scanner signal drift so it cannot mimic slow translation.

All frames are optimized simultaneously in PyTorch by maximizing Mattes mutual information against $\bar{I}_{\ell(k)}$, using coarse-to-fine Adam capture followed by full-resolution L-BFGS and a coupled Newton

polish. During resampling, the two-transform chain `[0000.mat, 0000_eddy.mat]` is composed and applied in a single pass using pure-PyTorch `float32` order-3 cardinal B-spline interpolation.

2. Curvature-normalized re-descending trajectory prior (Figure 1B, C). To suppress sub-voxel registration noise on low-SNR frames without flattening real head motion, we penalize the discrete second difference (acceleration) of the free head pose, $u_t = m_t - \frac{1}{2}(m_{t-1} + m_{t+1})$, where rotations are converted to millimetres on a 50 mm cortical radius. Because $u_t = 0$ for any constant-velocity motion, steady linear head drift is never penalized. The penalty added to the objective is:

$$\lambda\kappa \sum_t \rho(|u_t|), \qquad \rho(u) = \frac{\delta^2}{2}\log\Big(1 + \frac{u^2}{\delta^2}\Big), \qquad \rho'(u) = \frac{u}{1 + u^2/\delta^2}, \qquad \delta = 0.05 \text{ mm},\ \lambda = 0.03. \quad (2)$$

Two features make this prior work across diverse MRI sequences without manual tuning:

- **Curvature normalization (κ):** The raw sharpness of the mutual-information loss varies widely across modalities (for example, low-contrast ASL versus high-SNR BOLD). At each stage, the solver measures the median second derivative κ (in mm^{-2}) of the image metric with respect to translation. Multiplying λ by κ makes the weight $\lambda = 0.03$ dimensionless. In a motionless low-contrast ASL phantom, an unnormalized weight of 0.02 pulled the trajectory by 0.1 mm, whereas curvature normalization produced zero displacement.

- **Re-descending Cauchy influence function ($\rho'(u)$):** Standard convex penalties such as Charbonnier ($\rho(u) = \delta\sqrt{u^2 + \delta^2} - \delta^2$) or steep-origin L1 (`sparse`, knee $\delta/10$) exert a constant maximum pull $\rho'(u)$ even when $|u_t|$ is large, which drags real head jumps toward the mean of neighbouring frames. In contrast, the Cauchy derivative $\rho'(u)$ rises linearly for tiny wiggles below the knee $\delta = 0.05$ mm, peaks at $u = \delta$, and then *re-descends* toward zero ($\sim \delta^2/u$) for larger steps (Figure 1B). Thus sub-voxel noise wiggles are damped, while sudden head movements larger than a few tenths of a millimetre break free of the penalty and are estimated purely from the image data. Because the exact second derivative $\rho''(u)$ becomes negative above δ, our coupled Newton solver uses the positive-definite iteratively reweighted least-squares (IRLS) majoriser $(\rho'(u)/u)I_6$, solving the block-pentadiagonal system across active frames in pure-PyTorch `float32`.

3. Dual phantom-and-clinical acceptance gate. We did not choose the penalty form or weight by visual inspection. Instead, every candidate prior was required to pass two quantitative gates simultaneously:

1. **Ground-truth phantom gate ($n = 3$):** Synthetic 4D series with known planted motion trajectories across ASL (drift plus jitter), multi-shell DWI (motion plus a 0.4 mm per unit g eddy shift), and slow-flow MRI (four encoding states). A candidate prior was rejected if it increased FD error against ground truth by more than 10 % on any phantom, and it was required to reduce FD error on DWI.

2. **Clinical readout-floor gate ($n = 5$):** Unlike the phase-encode axis, the frequency-encode (readout) axis has high bandwidth and is physically immune to $\Delta B_0(t)$ and eddy-current shifts. Once design-locked contrast offsets $D_k\beta$ are removed, frame-to-frame steps along the readout axis represent a physical floor of true head jitter plus irreducible image noise. On five clinical series, a prior was rejected if it shrank the readout step median absolute deviation (MAD) by more than 0.02 mm (20 μm), degraded post-correction temporal standard deviation by more than 1 %, or changed mean FD by more than 2 %.

4. Variance ledger and frame censoring (FD vs. $\text{FD}_{\text{bandlimited}}$). For every series, the module `syntx.motion_decomposition` produces an auditable ledger in both the pose domain and the voxel intensity domain:

- **Pose step-variance shares:** Along each anatomical acquisition axis (readout and phase-encode), frame-to-frame step variance is partitioned into the percentage held by true rigid head pose (`0000.mat`, m_k), applied phase-encode distortion (`0000_eddy.mat`, E_k), and removed contrast-locked bias ($D_k\beta$).
- **Voxel intensity variance shares:** At foreground brain voxels, sequential Type I sums of squares (after factoring out γ_k) measure the percentage of temporal variance explained by the acquisition design (level means plus $l = 2, 4$ DWI directional terms), by the 6-DOF motion regressors after accounting for the design, and by the unexplained residual, both before (raw) and after motion correction.
- **Complementary role of FD and $\mathbf{FD_{bandlimited}}$:** Primary Framewise Displacement FD(m_k) is computed on a 50 mm radius from `0000.mat` alone, so it is already free of phase-encode $\Delta B_0(t)$ and eddy pseudomotion at any repetition time (TR). In fast acquisitions (TR ≤ 2.78 s, $T \geq 32$, such as 1.0 s BOLD), breathing also causes a real mechanical head nod in the sagittal plane (slice translation t_z and pitch r_x at 0.18–0.50 Hz). Because this nod is real movement of the skull, it remains in `0000.mat` for spatial resampling; however, summing its ± 0.20 mm oscillation on every 1 s frame inflates step-wise scrubbing rates. When a narrowband spectral peak (prominence ≥ 1.5) is detected in 0.18–0.50 Hz, $\mathrm{FD}_{\mathrm{bandlimited}}$ applies a cosine-tapered notch filter to the pose traces before differencing [2,4] so frame censoring targets sporadic head jerks rather than steady breathing.

5. Datasets. We evaluated the solver (`syntx 6.30.5`) on five clinical reference series: (1) resting-state BOLD ($T = 300$, TR 1.0 s, phase-encode $j-$), (2) 25-volume ASL with M0 calibration (TR 4.0 s), (3) 60-volume PCASL without M0 (30 tag/control pairs, TR 4.4 s), (4) 100-volume multi-shell DWI ($b = 1500/3000\,\mathrm{s/mm^2}$, dir-AP), and (5) 99-volume low-SNR multi-shell DWI ($b = 1500/3000\,\mathrm{s/mm^2}$, dir-PA). We also evaluated a 200-volume multi-shell DWI scan with severe eddy/$\Delta B_0(t)$ distortion (`sub-EAS024`) and an independent four-run task-fMRI dataset (OpenNeuro ds007693, sub-07, $4\times340 = 1360$ volumes, TR 1.5 s).

Results

1. Only the re-descending Cauchy prior at $\lambda = 0.03$ passes both the phantom and clinical gates (Figure 2). We first tested the intuition that small frame-to-frame second differences should be penalized most aggressively using the steep-origin `sparse` penalty and the convex Charbonnier penalty. On a 1D synthetic curve with Gaussian noise (Figure 1C), `sparse` at unit weight removed 71 % of sub-knee jitter energy compared with 54 % for Charbonnier and 53 % for Cauchy. However, on the 3D ground-truth phantoms (Figure 2A), both non-redescending penalties failed badly: at $\lambda = 0.10$, `sparse` increased FD error by +86 % on ASL and +55 % on slow-flow MRI, and Charbonnier increased FD error by +70 % and +50 %, because both penalties smoothed away true sub-voxel motion planted in the phantom trajectories.

Only the re-descending Cauchy form passed the phantom gate, reducing FD error by −14 % on ASL, −17 % on DWI, and −3 % on slow-flow at $\lambda = 0.03$ (and by −7 %, −27 %, and +3 % at $\lambda = 0.05$). The clinical readout-floor gate (Figure 2B, C) then separated the two weights: at $\lambda = 0.05$, Cauchy over-smoothed the B0-immune readout trace on the 100-volume DWI series, reducing the readout step MAD by 0.043 mm (from 0.081 to 0.038 mm, exceeding the 0.020 mm gate). At $\lambda = 0.03$, every clinical series stayed inside both acceptance bands (maximum readout step change 0.011 mm; image alignment unchanged within ≤ 0.1 %; FD change within −0.9 to +1.8 %). We therefore unified $\lambda = 0.03$ as the default across all one-level and multi-level designs.

2. Separating `[0000.mat, 0000_eddy.mat]` eliminates false phase-encode FD while sharpening 4D alignment (Table 2, Figure 3). Table 2 and Figure 3 report the complete pose and voxel variance decomposition across the five clinical series:

- **Multi-shell DWI ($T = 100$ and $T = 99$, plus `sub-EAS024` $T = 200$):** On the phase-encode axis, applied eddy and $\Delta B_0(t)$ shifts in `0000_eddy.mat` account for **87 % and 93 %** of the total frame-to-frame step variance, removed contrast-locked offsets $D_k\beta$ account for 7 % and 3 %, and true rigid head motion in `0000.mat` accounts for less than 1 %. On the B0-immune readout axis, $D_k\beta$ accounts for **68 % and 87 %** of the step variance (from $l = 2$ directional contrast shifts across shells), while true head motion accounts for 38 % and 9 %. Because earlier pipelines left the centered eddy and $\Delta B_0(t)$ phase-encode translations inside `0000.mat`, they reported mean FD of 0.60 mm (44 % of volumes > 0.50 mm) and 0.48 mm (36 % > 0.50 mm) on these two series. Transferring those phase-encode field shifts into `0000_eddy.mat` reduces mean rigid FD to 0.14 **mm (**0 % > 0.50 **mm) and** 0.15 **mm (**1 % > 0.50 **mm)** while applying the exact same total displacement during B-spline resampling. On the 200-volume `sub-EAS024` DWI scan, mean rigid FD similarly fell from 0.474 mm (71/200 volumes > 0.50 mm) to 0.173 mm (0/200 volumes > 0.50 mm) while keeping 0.680 mm RMS eddy shear and 0.303 mm RMS $\Delta B_0(t)$ shift in `0000_eddy.mat`. In the voxel domain, motion correction increases variance explained by the diffusion design from 94.3 to 96.2 % and from 90.8 to 95.6 %, cuts motion contamination after design from 0.57 to 0.45 % and from 1.76 to 0.66 %, and leaves the residual uncorrelated with FD ($r = -0.09$ and -0.04).

- **Resting-state BOLD ($T = 300$, TR 1.0 s):** Even without diffusion gradients, breathing-induced $\Delta B_0(t)$ field shifts along the phase-encode axis ($j-$) account for **53 %** of the raw phase-encode step variance. Moving $\delta_{\mathrm{pe},k}$ into `0000_eddy.mat` reduces the phase-encode step SD in `0000.mat` by 92.3 % in variance (from 0.140 mm down to 0.039 mm, matching the 0.047 mm readout noise floor) and lowers mean Power FD from 0.60 to 0.55 mm (38 % > 0.50 mm). The remaining step variance in `0000.mat` is dominated by real mechanical respiratory head nodding in the sagittal plane (slice t_z step SD 0.377 mm and pitch r_x step SD 0.193 mm oscillating at 0.48 Hz, prominence 1.94, holding 56.3 % of band power). Self-gated $\mathrm{FD}_{\mathrm{bandlimited}}$ notches this 0.355–0.50 Hz breathing band for censoring, reducing mean censoring FD to 0.28 **mm** and censored frames from 38 % to 11 %, while voxel motion variance falls from 36.41 to 25.62 %.

- **Arterial Spin Labeling ($T = 25$ with M0 and $T = 60$ PCASL):** Separating phase-encode $\Delta B_0(t)$ shifts into `0000_eddy.mat` accounts for **52 % and 56 %** of phase-encode step variance, lowering mean rigid FD from 0.51 to 0.47 mm on the 25-volume series and from 0.19 to 0.16 mm (2 % censored) on the 60-volume series. On the 25-volume series, tag/control+M0 design variance rises from 97.0 to 98.3 % and motion variance after design drops by half (0.35 to 0.18 %).

- **Execution speed:** Vectorizing the coupled Newton active-frame Cholesky solver and order-3 B-spline resampling in pure-PyTorch `float32` (and reusing the single-pass reference mask) reduced total wall-clock time across the five clinical series from 777 s (66 to 229 s per series) to 227 **s (**31 **to** 69 **s per series,** 3.4× **faster)**.

Discussion

Our results demonstrate why phase-encode distortion correction and head-motion reporting must be decoupled into two separate transform files (`[0000.mat, 0000_eddy.mat]`) rather than forced into one. On multi-shell DWI, 87 to 93 % of phase-encode step variance and 68 to 87 % of readout step

variance are driven by eddy currents, $\Delta B_0(t)$ field drift, and shell/direction contrast rather than movement of the skull. If an algorithm forces the applied phase-encode translation to be temporally smooth, real frame-to-frame $\Delta B_0(t)$ voxel shifts go uncorrected during resampling and blur the shells; if an algorithm keeps those shifts inside a single rigid transform, the resampled image is sharp but the reported FD trace reflects field distortion rather than patient motion. Writing the rigid skull pose to `0000.mat` and the phase-encode eddy plus $\Delta B_0(t)$ shifts to `0000_eddy.mat` solves both problems across all time-series modalities.

Equally important, our phantom and clinical gates falsified two plausible regularization hypotheses. First, penalties with steep origin pull (`sparse`) or constant large-step pull (Charbonnier) removed the most noise on a 1D toy curve but degraded 3D phantom accuracy by 50 to 86 % because they smoothed real motion steps. Second, amplitude shrinkage toward the identity transform (`pose_shrinkage`) lowered clinical FD by 11 to 41 % within the readout floor but increased ASL drift-phantom error two- to three-fold by pulling drifting heads toward zero. Only a curvature-normalized, re-descending second-difference prior ($\lambda = 0.03$) preserved both linear drift and real jumps while damping sub-voxel noise.

Limitations. (1) Our 3D phantom generators plant sub-knee jitter in the ground-truth motion trajectory rather than synthesizing independent thermal noise across image repetitions; a dedicated k-space/image-noise phantom gate is needed next. (2) On the 60-volume PCASL series without an M0 volume, the tag-control design explains only 1.1 % of voxel variance while motion regressors explain 53.75 %, reflecting strong spatial coupling between perfusion contrast and slice-wise signal changes that pose regression alone does not separate. (3) Censoring thresholds (0.50 mm in `antsxqc`) have not yet been benchmarked against downstream test-retest connectivity or tractography reproducibility on these five series. (4) An alternative engine that projects design columns inside the image metric (`engine="projected"`) over-reports B0-immune readout jitter on real multi-shell DWI (0.145 vs. 0.080 mm) and is therefore not used by default. (5) Any true physical head motion that oscillates at the exact frequency of the acquisition design (for example, Nyquist tag-control alternation) lies in the column space of D_k and is indistinguishable from $D_k\beta$.

Novelty Assessment

What builds on prior work. Using mutual information to reduce sensitivity to activation contrast follows Freire and Mangin [1]. Modeling diffusion eddy currents along the phase-encode axis as a gradient-linear affine with Jacobian intensity modulation follows Andersson and Sotiropoulos [3]. Filtering respiratory oscillations from framewise displacement before scrubbing follows Power et al. [2] and Fair et al. [4].

What is new in this contribution.

1. *Universal `[0000.mat, 0000_eddy.mat]` split and $D_k\beta$ contrast removal across BOLD, ASL, DWI, and slow-flow MRI:* A single joint solver separates true 3D rigid head pose (`0000.mat`), applied phase-encode eddy and Wiener-shrunk $\Delta B_0(t)$ field shifts (`0000_eddy.mat`, resampled together in one pure-PyTorch `float32` B-spline pass), and unapplied contrast-locked pose bias ($D_k\beta$).

2. *Curvature-normalized re-descending Cauchy prior ($\lambda = 0.03$) with active-frame IRLS Newton optimization:* Normalizing the second-difference penalty by stage-wise median translation curvature κ makes $\lambda = 0.03$ invariant to image contrast, while the re-descending Cauchy pull damps sub-knee (< 0.05 mm) wiggles without shrinking drift or jumps.

3. *Dual ground-truth phantom and B0-immune readout-floor acceptance gate:* Combining synthetic drifting phantoms with the physical requirement that regularization must not shrink steps along the distortion-immune readout axis ($< 20\,\mu$m) falsified convex L1/Charbonnier priors, amplitude shrinkage, and over-weighted Cauchy ($\lambda = 0.05$).

4. *Closed-form pose and voxel variance ledger (`syntx.motion_decomposition`):* Quantifies the exact percentage of step variance and voxel intensity variance attributable to rigid head motion, applied phase-encode distortion, and removed contrast bias in every scan.

Conclusion

Across BOLD, ASL, multi-shell diffusion, and slow-flow MRI, a single joint objective with a $\lambda = 0.03$ redescending Cauchy prior separates applied transforms into `[0000.mat, 0000_eddy.mat]` and removes contrast-locked bias $D_k\beta$—producing sharp 4D volumes and uncontaminated head-motion FD traces 3.4× faster.

References

[1] Freire L, Mangin JF. NeuroImage 2001. [2] Power JD et al. PNAS 2019. [3] Andersson JLR, Sotiropoulos SN. NeuroImage 2016. [4] Fair DA et al. NeuroImage 2020.

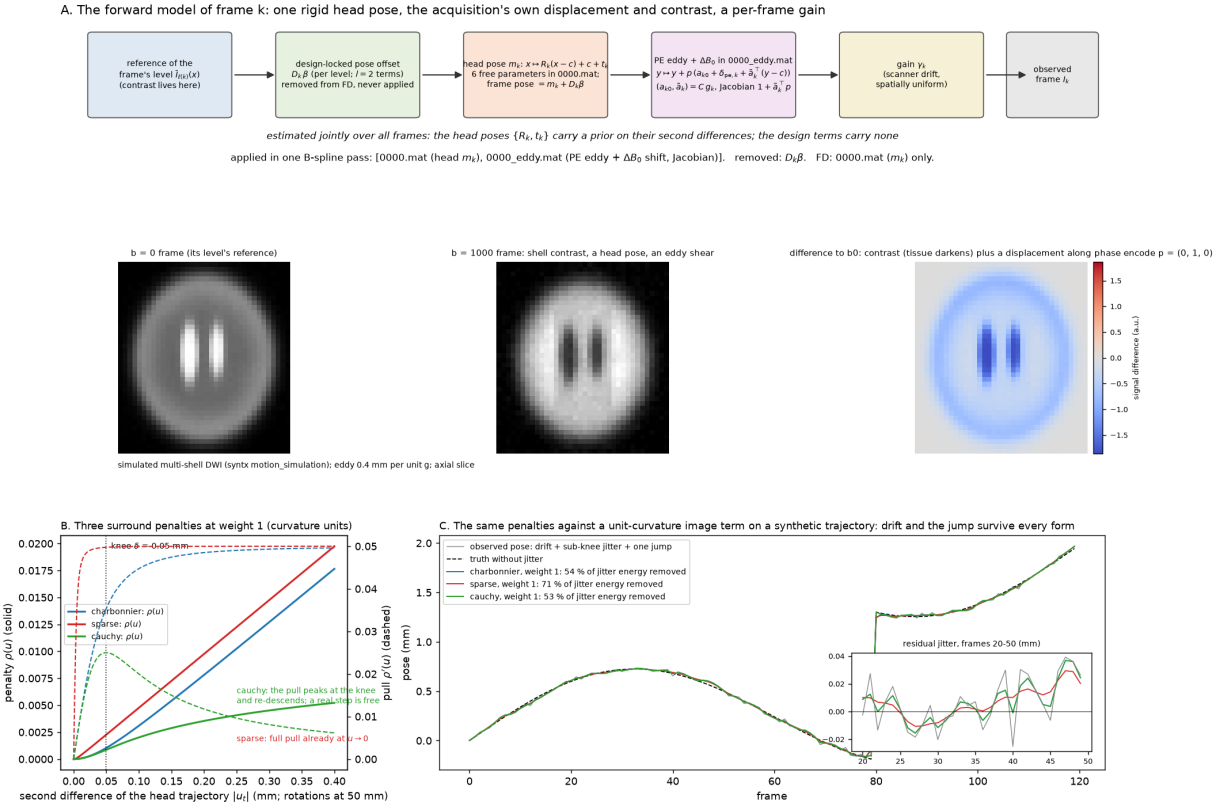


Figure 1: The forward generative model separates physical displacements from contrast bias, and the re-descending Cauchy penalty preserves real motion steps while damping sub-voxel noise. (A) **Forward model of frame k (Eq. 1):** Each frame is compared against the mean image of its own acquisition level $\bar{I}_{\ell(k)}(x)$. Contrast-locked pose offsets $D_k\beta$ are subtracted and never resampled. True 3D rigid head pose m_k is stored in `0000.mat`, while phase-encode eddy shear/translation Cg_k and bulk $\Delta B_0(t)$ field shifts $\delta_{\mathrm{pe},k}$ (with Jacobian $1 + \tilde{a}_k^\top p$) are stored in `0000_eddy.mat`; both transforms are applied together in one B-spline interpolation pass. Middle row: simulated multi-shell DWI axial slices ($b = 0$, $b = 1000\,\mathrm{s/mm}^2$, and their difference), illustrating simultaneous tissue darkening (contrast) and anterior-posterior rim displacement along the phase-encode axis $p = (0, 1, 0)$. (B) **Penalty $\rho(u)$ and restoring pull $\rho'(u)$:** Plotted at unit curvature weight against the trajectory second difference $|u_t|$ (in mm). Convex Charbonnier (blue) and steep-origin `sparse` (red) maintain a strong pull at large $|u_t|$, whereas the Cauchy pull (green) peaks at the knee $\delta = 0.05\,\mathrm{mm}$ and decays toward zero so large steps are not shrunk. (C) **1D trajectory denoising test ($T = 120$ frames):** On a synthetic curve combining slow drift, sub-knee Gaussian jitter, and a 1.5 mm step jump at frame 80, all three second-difference penalties preserve linear drift and the 1.5 mm jump, while the inset (frames 20–50) shows how each form damps sub-knee oscillations.

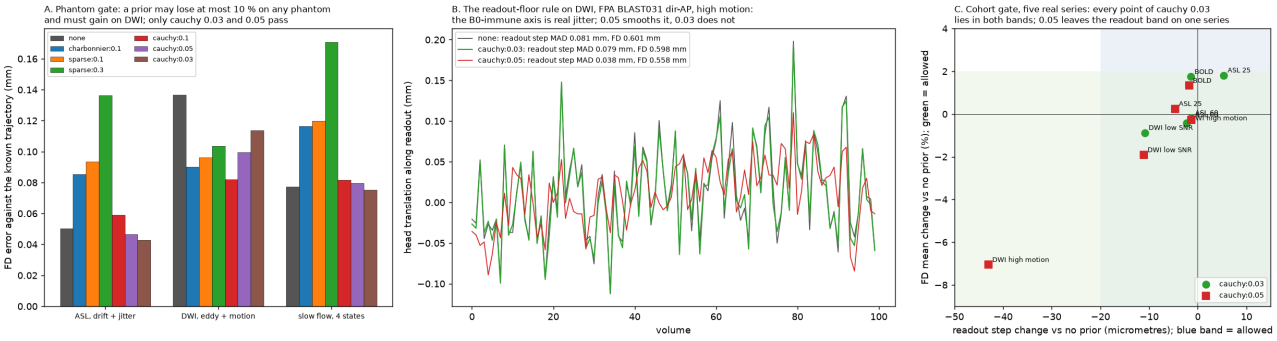


Figure 2: Dual acceptance gate: only the re-descending Cauchy prior at $\lambda = 0.03$ passes both the ground-truth phantom gate and the clinical readout-floor gate. (A) **Ground-truth phantom gate ($n = 3$):** Framewise displacement (FD) error in mm against the known planted trajectory on synthetic ASL (drift + jitter), multi-shell DWI (motion + eddy), and 4-state slow-flow MRI across seven prior settings (`form:weight`). Non-redescending priors (`sparse:0.1`, `charbonnier:0.1`) increase FD error by 50 to 86 % because they smooth away true sub-knee motion; only `cauchy:0.03` and `cauchy:0.05` reduce DWI error while staying within the +10 % ceiling on all phantoms. (B) **B0-immune readout-floor test on clinical DWI ($T = 100$):** Because the readout axis is immune to $\Delta B_0(t)$ and eddy distortion, its post-$D_k\beta$ step MAD (0.081 mm with no prior, grey) represents true head jitter plus noise floor. Raising the Cauchy weight to $\lambda = 0.05$ (red) over-smooths this trace (MAD drops to 0.038 mm, $-43\,\mu$m), whereas $\lambda = 0.03$ (green, MAD 0.079 mm, $-2\,\mu$m) preserves it. (C) **Five-series clinical gate summary:** Readout step MAD change (μm, allowed band $\pm 20\,\mu$m shaded blue) versus mean FD change (%, allowed band $\leq +2$ % shaded green) across all five clinical series. Every series under `cauchy:0.03` (green circles) lies inside both acceptance bands.

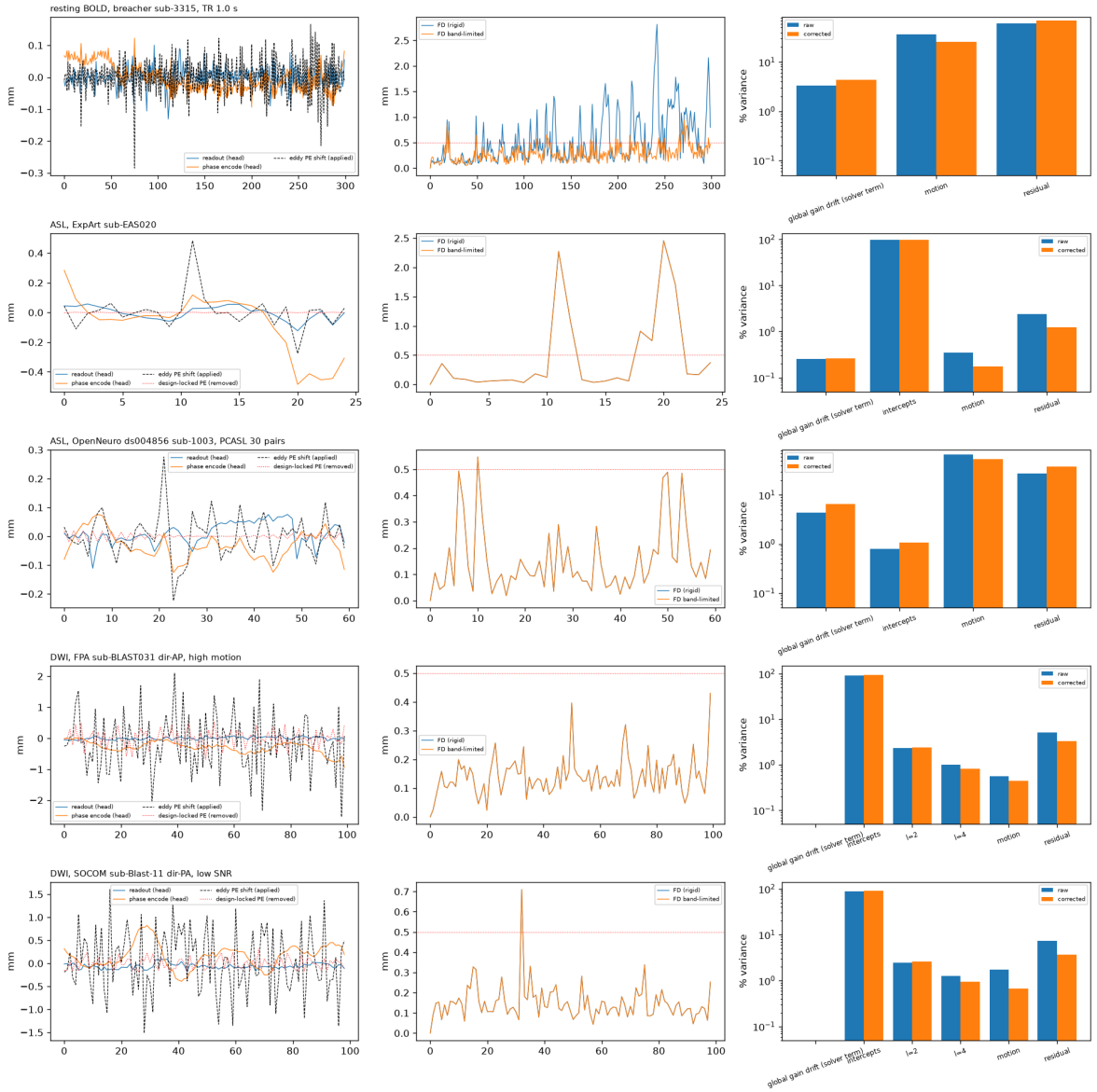


Figure 3: Pose and voxel variance decomposition across the five clinical series (`syntx 6.30.5`). Each row shows one clinical series: resting BOLD ($T = 300$), 25-volume ASL with M0, 60-volume PCASL, 100-volume multi-shell DWI, and 99-volume low-SNR multi-shell DWI. **Left column (Pose decomposition in mm vs. frame):** Translation along the B0-immune readout axis (blue solid) and phase-encode axis (orange solid) from the pure rigid transform `0000.mat`, overlaid with the applied phase-encode eddy+$\Delta B_0(t)$ shift in `0000_eddy.mat` (black dashed) and the removed design-locked contrast bias $D_k\beta$ (red dotted). Note how `0000_eddy.mat` captures the large rapid phase-encode oscillations across all five series while `0000.mat` tracks smooth, physically plausible skull motion. **Middle column (Framewise displacement in mm vs. frame):** Primary rigid FD(m_k) from `0000.mat` (blue) and censoring $\mathrm{FD}_{\mathrm{bandlimited}}$ (orange) relative to the 0.50 mm `antsxqc` scrubbing threshold (red dotted line). On resting BOLD (top row), notching the 0.48 Hz sagittal breathing nod reduces censored frames from 38 % to 11 %; on both DWI series (bottom two rows), separating phase-encode distortion into `0000_eddy.mat` keeps rigid FD below 0.50 mm on 99–100 % of volumes. **Right column (Voxel temporal variance explained, log scale):** Percentage of foreground voxel variance explained by scanner gain drift, acquisition design terms (level intercepts and $l = 2, 4$ DWI spherical harmonics), motion regressors after design, and residual, before (blue) and after (orange) correction. Motion-related voxel variance decreases after correction in 5 of 5 series.

Table 2: Quantitative pose and image variance ledger across the five clinical reference series (`syntx 6.30.5`). FD mean / max: Power framewise displacement (50 mm radius) computed from the pure rigid head transform `0000.mat` (m_k). **Censored (%):** Percentage of frames exceeding the 0.50 mm `antsxqc` scrubbing threshold on rigid FD(m_k) / on the censoring trace ($\mathrm{FD}_{\mathrm{bandlimited}}$ for BOLD and ASL, rigid FD for DWI). **Readout and PE h/e/d (%):** Percentage of frame-to-frame step variance along the readout and phase-encode axes held by rigid head motion (**h**, `0000.mat`) / applied phase-encode eddy+$\Delta B_0(t)$ shift (**e**, `0000_eddy.mat`) / removed design-locked contrast bias (**d**, $D_k\beta$). (Component shares need not sum to 100 % when components covary slightly over finite runs.) **Design, Motion, Residual (%):** Percentage of foreground voxel temporal variance explained by the acquisition design, by motion regressors after accounting for the design, and by the unexplained residual, shown as `raw / corrected`. r**:** Correlation between per-frame residual RMS and FD after correction. Rows are generated automatically from `docs/reports/ismrm_cases_6_30_5_2026-10-29.json` by `benchmarks/ismrm_table.py`.

case	T	FD mean / max (mm)	censored (%)	readout h/e/d (%)	PE h/e/d (%)	design (%)	motion (%)	residual (%)	r	time
resting BOLD, TR 1.0 s	300	0.55 / 2.81	38 / 11	100 / 0 / 0	8 / 53 / 0	0.0 / 0.0	36.41 / 25.62	60.3 / 70.0	0.60	33 s
ASL, control/label + M0, 25 vol	25	0.47 / 2.46	24 / 24	101 / 0 / 2	13 / 52 / 0	97.0 / 98.3	0.35 / 0.18	2.4 / 1.2	0.69	32 s
ASL, control/label, 60 vol	60	0.16 / 0.55	2 / 2	86 / 0 / 3	6 / 56 / 3	0.8 / 1.1	67.23 / 53.75	27.6 / 38.5	0.61	31 s
DWI multi-shell, high motion	100	0.14 / 0.43	0 / 0	38 / 0 / 68	0 / 87 / 7	94.3 / 96.2	0.57 / 0.45	5.1 / 3.4	-0.09	69 s
DWI multi-shell, low SNR	99	0.15 / 0.71	1 / 1	9 / 0 / 87	0 / 93 / 3	90.8 / 95.6	1.76 / 0.66	7.4 / 3.7	-0.04	62 s